

ISO 286 fits chart: H7/g6, H7/h6, H7/p6 and deviations

1. The 8 fits to know, and when to use them

The **hole-basis system** (H hole, you choose the shaft) is by far the most used: one reamer or hole tolerance across the range, and the shaft, easier to machine precisely, carries the function. These eight pairs cover almost all general engineering.

Fit	Type	Typical uses
H11/d11	Very loose	Rough parts, hinge pins, deliberately large clearance
H9/d9	Free running	Low-precision bearings, idler pulleys, high speed without precision
H8/f7	Running	Lubricated plain bearings, shafts rotating in a bush
H7/g6	Sliding (close)	Precise guiding with no noticeable play: sliding gears, slides, cylinder pistons
H7/h6	Locational clearance	Hand-removable locations: flanges, spacers, locating parts
H7/k6	Transition	Hubs, pulleys and couplings fitted with a mallet, with a key
H7/n6	Transition to tight	Light press fit: bushes, sleeves, parts rarely taken apart
H7/p6	Interference	Press fit: friction bushes, dowel pins in steel

For **rolling bearings**, don't pick from this table: the shaft and housing tolerance depends on the load (rotating or stationary) and its magnitude. Bearing manufacturers publish their own recommendations — they take precedence.

2. Deviation chart: H7 and the most common shafts

Upper / lower deviations in micrometres, by nominal size range (over ... up to and including ...). Example: a Ø 25 g6 shaft measures between 24.980 and 24.993 mm.

Nominal Ø (mm)	H7	f7	g6	h6	k6	n6	p6
≤ 3	+10 / 0	-6 / -16	-2 / -8	0 / -6	+6 / 0	+10 / +4	+12 / +6
3 - 6	+12 / 0	-10 / -22	-4 / -12	0 / -8	+9 / +1	+16 / +8	+20 / +12
6 - 10	+15 / 0	-13 / -28	-5 / -14	0 / -9	+10 / +1	+19 / +10	+24 / +15
10 - 18	+18 / 0	-16 / -34	-6 / -17	0 / -11	+12 / +1	+23 / +12	+29 / +18
18 - 30	+21 / 0	-20 / -41	-7 / -20	0 / -13	+15 / +2	+28 / +15	+35 / +22
30 - 50	+25 / 0	-25 / -50	-9 / -25	0 / -16	+18 / +2	+33 / +17	+42 / +26
50 - 80	+30 / 0	-30 / -60	-10 / -29	0 / -19	+21 / +2	+39 / +20	+51 / +32
80 - 120	+35 / 0	-36 / -71	-12 / -34	0 / -22	+25 / +3	+45 / +23	+59 / +37
120 - 180	+40 / 0	-43 / -83	-14 / -39	0 / -25	+28 / +3	+52 / +27	+68 / +43
180 - 250	+46 / 0	-50 / -96	-15 / -44	0 / -29	+33 / +4	+60 / +31	+79 / +50
250 - 315	+52 / 0	-56 / -108	-17 / -49	0 / -32	+36 / +4	+66 / +34	+88 / +56
315 - 400	+57 / 0	-62 / -119	-18 / -54	0 / -36	+40 / +4	+73 / +37	+98 / +62
400 - 500	+63 / 0	-68 / -131	-20 / -60	0 / -40	+45 / +5	+80 / +40	+108 / +68

Values from the ISO 286-2 tables: IT tolerance grades and fundamental deviations from ISO 286-1.

Reference values taken from the article <https://www.baouch.fr/en/blog/ajustements-iso-286-tableau-h7-g6/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.